

AHMED TAREK EMAM

+20 127 652 6064 | engatemam@gmail.com | [linkedin.com/in/engatemam](https://www.linkedin.com/in/engatemam) | Ismailia, Egypt

PROFESSIONAL SUMMARY

Dedicated Senior Electrical Engineering student (Electronics & Communications) with practical expertise in RF/Microwave design, Software-Defined Radio (SDR), and Telecommunications. Backed by solid foundations in Cloud Architecture, Network Security, and Generative AI. Proven ability to simulate and implement complex engineering systems (HFSS, CST, MATLAB, Cisco Packet Tracer). Seeking to leverage cross-functional skills in an internship or entry-level role.

EDUCATION

Suez Canal University

Bachelor of Electrical Engineering (Electronics & Communications Dept.)

– Cumulative GPA: 3.17 / 4.00 (Very Good)

Ismailia, Egypt

Exp. Graduation: Sep. 2026

EXPERIENCE & TRAINING

Cyber Security Trainee

Information Technology Institute (ITI)

– Gained hands-on expertise in Network Security, Ethical Hacking, Linux, and Python.

July 2025 — Aug. 2025

Ismailia, Egypt

Cloud Architecture Trainee

Information Technology Institute (ITI)

– Mastered Cloud Computing & DevOps methodologies, including Docker, Kubernetes, and AWS infrastructure.

July 2025 — Aug. 2025

Ismailia, Egypt

Gen AI for Beginners Trainee (Powered by NVIDIA)

Information Technology Institute (ITI)

– Completed a 35-hour intensive camp focused on Generative AI, RAG techniques, and LLM application development.

Sep. 2025 — Oct. 2025

Remote

TECHNICAL PROJECTS

RF-Based Detection & SDR Scanning | Linux, RTL-SDR, GQRX

- Constructed a real-time spectrum analyzer using RTL-SDR V4 and GQRX on Linux to intercept live radio signals.
- Captured and analyzed GSM Uplink signals targeting 902.2 MHz, bridging theoretical DSP with ML localization.

Compact Hexagonal Dual-Band Antenna | HFSS, FR-4 Substrate, VNA

- Designed and simulated a wearable WBAN antenna operating at 2.4 GHz and 5.0 GHz using Ansys HFSS.
- Validated bio-safety ($SAR < 1.6 \text{ W/kg}$) and conducted physical prototype testing using VNA at NTI.

Enterprise Network Architecture Design | Cisco Packet Tracer, CCNA

- Engineered a multi-floor topology for the Engineering Faculty, integrating RIP routing, DHCP, and SSH security.

FMCW Radar & Analog Comm. Systems | MATLAB, DSP

- Simulated a 77 GHz FMCW Radar utilizing FFT for precise target range and velocity tracking, alongside CFAR detection.
- Modeled a Superheterodyne receiver integrating SSB/NBFM modulation and FDM to analyze signal fidelity.

TECHNICAL SKILLS

Languages & AI: C++, Python, MATLAB, Generative AI (LLMs, RAG, Prompt Engineering)

RF & Telecom: SDR, GSM Uplink, Antenna Design, Microwave Engineering, Radar Systems (FMCW)

Tools & Software: Ansys HFSS, CST Studio, GQRX, Cisco Packet Tracer, Git, Linux (Ubuntu), AWS, Docker

CERTIFICATIONS

AI & Data: NVIDIA (AI for All, Building LLM Apps, RAG) | Cisco (Data Analytics) | IEEE SCU (AI Track)

Cloud & Networks: Huawei (HCCDA-Tech) | AWS (Cloud Foundations) | ITI (Linux) | Creativa (CCNA)

LANGUAGES

Arabic: Native

English: Intermediate